

Weichen Lyu

Mobile: (+86) 18101038161 • Email: lvwc25@mails.tsinghua.edu.cn

WeChat: [echo_hello_there](#) • Homepage: matthew-lyu.github.io

Education

Tsinghua University, M.Eng. Student

Sep. 2025 – Present

Artificial Intelligence

GPA: 3.8/4.0 Expected graduation: 2028

Department of Automation

Supervisor: Prof. [Yueqi Duan](#)

Research Topic: 3D Vision and Embodied AI

Research Focus: Specializing in 3D vision technologies and applications

Sun Yat-sen University, Undergraduate

Sep. 2021 – Jun. 2025

B.Eng. in Intelligent Science and Technology

GPA: 3.8/4.0

School of Intelligent Systems Engineering

Core Course: Deep Learning (91/100); Fundamentals of Autonomous Driving (99/100); Optimization Theory and Methods (90/100); Multi-Agent Swarm Control Technologies (99/100); Natural Language Processing (90/100).

Publications and Preprints

* Equal contribution. † Corresponding author.

- [1] **PhySGA: A Physics-Space Grounded Agentic Framework for Generalizable and Interactive 3D Composition.**
Weichen Lyu*, Chong Xia*, Mingyu Dong, Long Xu, Shengjun Zhang, Yueqi Duan†.
Submitted to ICLR 2027. Co-first author. [Project page](#)
- [2] **ReplicateAnyScene: Zero-Shot Video-to-3D Composition via Textual-Visual-Spatial Alignment.**
Mingyu Dong*, Chong Xia*, Mingyuan Jia, **Weichen Lyu**, Long Xu, Zheng Zhu, Yueqi Duan†.
arXiv preprint, arXiv:2604.10789, 2026. [Paper](#) [Project page](#)
- [3] **A Novel End-to-End 3-D Residual and Attention Enhancement Joint Few-Shot Learning Model for Huntington Clinical Assessment.**
Jiehui Huang, Lishan Lin, Jiaying Lin, Jun Zhou, Baiqiao Yin, **Weichen Lyu**, Fengjuan Su, Kang Yuan, Mansi Cai, Yucheng Li, Haifan Kong, Jean-Marc Burgunder, Zhong Pei, Calvin Yu-Chian Chen†.
IEEE Transactions on Instrumentation and Measurement, vol. 74, pp. 1–11, 2025. [Paper](#)

Research Experience

Research Assistant, The Chinese University of Hong Kong, Shenzhen

May. – Sep. 2025

Shenzhen Key Laboratory of Cross-modal Cognitive Computing (C3 Lab)

SpeechLab, School of Data Science

Supervisor: Prof. [Zhizheng Wu](#)

Research focus: Text-to-Speech (TTS) post-training.

Honors and Awards

Outstanding Student Scholarship, Sun Yat-sen University

2021–2022

Outstanding Student Scholarship, Sun Yat-sen University

2022–2023

Special Scholarship (Professional Competence Award), Sun Yat-sen University

2022–2023